1 Definitions

Moiety: A subgroup of a larger molecule.
Conserved Moiety: A subgroup whose total amount remains constant during interconversion through a sequence of reactions.

1.1 Examples of Conserved Moieties

- Phosphate groups (ATP/ADP/AMP)
- Proteins in phosphorylation cycles
- NAD/NADH
- Enzyme/enzyme-substrate (E/ES)
- Coenzyme A

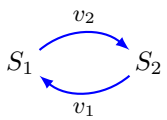
1.2 Time Scale Separation

Conservation is an **approximation**:

- Interconversions: **fast** (seconds–minutes)
- Synthesis/degradation: **slow** (hours–days)

Over short time scales, total moiety $\approx$ constant.

2 Simple Conserved Cycle



2.1 System Equations

$$\frac{ds_1}{dt} = v_1 - v_2$$
$$\frac{ds_2}{dt} = v_2 - v_1$$

Key observation:

$$\frac{ds_1}{dt} = -\frac{ds_2}{dt}$$

Therefore: $s_1 + s_2 = T$ (constant)

2.2 Reduced System

Only need **one** ODE + **one** algebraic equation:

$$s_2 = T - s_1$$
$$\frac{ds_1}{dt} = v_1 - v_2$$

T = total moiety (from initial conditions)

3 Detecting Conservation Laws

3.1 Key Insight

Row dependencies in N $\Leftrightarrow$ Conservation laws

For simple cycle:

$$\mathbf{N} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Row 2 $= -1\times$ Row 1 $\Rightarrow$ dependent rows!

3.2 Counting

Quantity	Formula
Total species	m
Rank of $\mathbf{N}$	$r = \text{rank}(\mathbf{N})$
Independent species	r
Dependent species	$m - r$
Conservation laws	$m - r$

3.3 Three-Species Cycle Example

$$\mathbf{N} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Sum of rows = 0, so:

$$\frac{ds_1}{dt} + \frac{ds_2}{dt} + \frac{ds_3}{dt} = 0$$

$\Rightarrow$ Conservation: $s_1 + s_2 + s_3 = T$

4 Row Reduction Method

4.1 Algorithm

Step 1: Form augmented matrix $[\mathbf{N} \mid \mathbf{I}]$
Step 2: Row reduce $\mathbf{N}$ to echelon form, applying same operations to $\mathbf{I}$
Step 3: Zero rows in reduced $\mathbf{N} \Rightarrow$ corresponding rows in transformed $\mathbf{I}$ are conservation laws

4.2 Mathematical Form

After row reduction:

$$\begin{bmatrix} \mathbf{M} \\ \mathbf{0} \end{bmatrix} \mathbf{v} = \begin{bmatrix} \mathbf{X} \\ \mathbf{Y} \end{bmatrix} \frac{d\mathbf{S}}{dt}$$

The $\mathbf{Y}$ partition gives conservation laws:

$$\mathbf{Y}\frac{d\mathbf{S}}{dt} = \mathbf{0}$$

4.3 Worked Example

Given linked cycles (E, ES, S_1, S_2):

$$\mathbf{N} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

After row reduction, extract bottom rows:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Conservation laws:

$$s_2 + es + s_1 = T_1$$
$$es + e = T_2$$

5 Null Space Connection

The conservation matrix $\boldsymbol{\gamma}$ is the **null space** of $\mathbf{N}^T$:

$$\mathbf{N}^T\boldsymbol{\gamma}^T = \mathbf{0}$$

Equivalently: $\boldsymbol{\gamma}$ is the **left null space** of $\mathbf{N}$:

$$\boldsymbol{\gamma}\mathbf{N} = \mathbf{0}$$

5.1 Verification Test

To check if conservation laws are correct:

$$\mathbf{N}^T \cdot (\text{conservation vectors}) = \mathbf{0}$$

5.2 Software Commands

Tool	Command
MATLAB	`null(N', 'r')`
Scilab	`rref(kernel(N'))'`
Python/SymPy	`Matrix(N).T.nullspace()`
Mathematica	`NullSpace[Transpose[N]]`

6 Matrix Partitioning

6.1 Stoichiometry Matrix Structure

Reorder rows: independent ($\mathbf{S}_I$) on top, dependent ($\mathbf{S}_D$) on bottom:

$$\mathbf{N} = \begin{bmatrix} \mathbf{N}_R \\ \mathbf{N}_0 \end{bmatrix}$$

- $\mathbf{N}_R$: reduced matrix (full rank, $m_0 \times n$)
- $\mathbf{N}_0$: dependent rows

6.2 Link-Zero Matrix $\mathbf{L}_0$

$\mathbf{N}_0$ is a linear combination of $\mathbf{N}_R$:

$$\mathbf{N}_0 = \mathbf{L}_0 \mathbf{N}_R$$

Dimensions: $\mathbf{L}_0$ is $(m - m_0) \times m_0$

6.3 Link Matrix L

$$\mathbf{L} = \begin{bmatrix} \mathbf{I} \\ \mathbf{L}_0 \end{bmatrix}$$

Key relationship:

$$\mathbf{N} = \mathbf{L} \cdot \mathbf{N}_R$$

If no conservation: $\mathbf{L} = \mathbf{I}$

6.4 Conservation Matrix

$$\boldsymbol{\gamma} = \begin{bmatrix} -\mathbf{L}_0 & \mathbf{I} \end{bmatrix}$$

Conservation law:

$$\boldsymbol{\gamma}\mathbf{S} = \mathbf{T}$$

Or explicitly:

$$\begin{bmatrix} -\mathbf{L}_0 & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{S}_I \\ \mathbf{S}_D \end{bmatrix} = \mathbf{T}$$

6.5 Computing Dependent Species

$$\mathbf{S}_D = \mathbf{L}_0 \mathbf{S}_I + \mathbf{T}$$

where $\mathbf{T} = \mathbf{S}_D(0) - \mathbf{L}_0 \mathbf{S}_I(0)$

7 Reduced System

7.1 Full vs Reduced

Full system: m ODEs

$$\frac{d\mathbf{S}}{dt} = \mathbf{N}\mathbf{v}$$

Reduced system: m_0 ODEs + $(m - m_0)$ algebraic

$$\frac{d\mathbf{S}_I}{dt} = \mathbf{N}_R \mathbf{v}$$

$$\mathbf{S}_D = \mathbf{L}_0 \mathbf{S}_I + \mathbf{T}$$

7.2 Why Reduce?

1. Fewer ODEs to integrate
2. Avoids numerical drift
3. Required for Jacobian inversion

8 Jacobian Implications

Problem: Conservation laws create **singular Jacobian**!
Row dependencies in $\mathbf{N}$ $\Rightarrow$ row dependencies in $\mathbf{J}$

For simple cycle with $v_1 = k_1 s_1$, $v_2 = k_2 s_2$:

$$\mathbf{J} = \begin{bmatrix} -k_1 & k_2 \\ k_1 & -k_2 \end{bmatrix}$$

$\det(\mathbf{J}) = 0$ — cannot invert!

Solution: Work with the **reduced system** to get an invertible Jacobian.

9 Numerical Methods

9.1 Method Comparison

Method	Stability	Speed
Row reduction	Poor	Fast
SVD	Excellent	Slow
QR factorization	Excellent	Medium

9.2 SVD Method

Factor $\mathbf{N}^T = \mathbf{U}\mathbf{S}\mathbf{V}^T$

Null space = bottom rows of $\mathbf{V}^T$ (where $\sigma_i = 0$)

Apply `rref()` for rational basis.

9.3 QR Method

Factor $\mathbf{N}^T\mathbf{P} = \mathbf{Q}\mathbf{R}$

Extract $\mathbf{L}_0$ from reduced $\mathbf{R}$:

$$\mathbf{L}_0 = \mathbf{R}_{12}^T$$

after row reducing top portion.

9.4 Best Practice

Row ordering tip: Place **shared species** (complexes) at **bottom** of $\mathbf{N}$ to get positive conservation law coefficients.

10 Multicompartment Systems

Conservation is of MASS, not concentration!
For cross-compartment conservation:

$$\sum_i V_i s_i = T$$

where V_i = volume of compartment i

11 Summary of Key Equations

Relation	Equation
Full system	$\mathbf{N} = \mathbf{L}\mathbf{N}_R$
Link-zero	$\mathbf{N}_0 = \mathbf{L}_0\mathbf{N}_R$
Conservation matrix	$\boldsymbol{\gamma} = [-\mathbf{L}_0 \mid \mathbf{I}]$
Null space	$\mathbf{N}^T\boldsymbol{\gamma}^T = \mathbf{0}$
Conservation law	$\boldsymbol{\gamma}\mathbf{S} = \mathbf{T}$
Dependent species	$\mathbf{S}_D = \mathbf{L}_0\mathbf{S}_I + \mathbf{T}$

12 Quick Reference: Tellurium

```
import tellurium as te
r = te.loada('''
  S1 -> S2; k1*S1
  S2 -> S1; k2*S2
  S1 = 10; k1 = 0.1; k2 = 0.2
''')
r.conservedMoietyAnalysis = True
print(r.getConservationMatrix())
print(r.getLinkMatrix())
```

Based on: H.M. Sauro, "Systems Biology: Introduction to Pathway Modeling", Created: March 19, 2026